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DETAIL SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIALS, POLYETHERSULFONE

REACTIVATED AFTER 5 JUNE 2026 AND MAY BE USED FOR NEW AND EXISTING DESIGN AND ACQUISITIONS.

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers polyethersulfone (PES) thermoplastic materials suitable for injection molding or extrusion.

1.2 Classification. The PES thermoplastic materials shall be of the following types and classes (see 6.2):

Type I - Nonreinforced

Class 1 - Injection molding resin

Grade A - Melt flow = 43 ± 4 grams/10 min
Grade B - Melt flow = 14.5 ± 1.5 grams/10 min

Class 2 - Extrusion resin

Grade A - Melt flow = 3.4 ± 0.4 grams/10 min

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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Type II - Glass fiber reinforced (GFR)

Class 1 - 15% GFR
Class 2 - 20% GFR
Class 3 - 30% GFR
Class 4 - 40% GFR
Class 5 - 15% PTFE/30% GFR

Type III - Carbon fiber reinforced (CFR)

Class 1 - 30& CFR
Class 2 - 15% PTFE/30% CFR

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

CFR TITLE 14 - Aeronautics and Space

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FEDERAL SPECIFICATIONS

PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

ASTM INTERNATIONAL

ASTM D149	-	Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
ASTM D150	-	Standard Test Methods for AC Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulation
ASTM D256	-	Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics
ASTM D257	-	Standard Test Methods for DC Resistance or Conductance of Insulating Materials
ASTM D570	-	Standard Test Method for Water Absorption of Plastics
ASTM D638	-	Standard Test Method for Tensile Properties of Plastics
ASTM D648	-	Standard Test Method for Deflection Temperature of Plastics Under Flexural Load in the Edgewise Position
ASTM D696	-	Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30°C and 30°C with a Vitreous Silica Dilatometer
ASTM D785	-	Standard Test Method for Rockwell Hardness of Plastics and Electrical Insulating Materials
ASTM D790	-	Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D792	-	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D955	-	Standard Test Method of Measuring Shrinkage from Mold Dimensions of Thermoplastics
ASTM D1238	-	Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D2863	-	Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)

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ASTM D3029	-	Standard Test Method for Impact Resistance of Monolithic Polycarbonate Sheet by Means of a Falling Weight
ASTM D3171	-	Standard Test Methods for Constituent Content of Composite Materials
ASTM D3418	-	Standard Test Method for Transition Temperatures and Enthalpies of Fusion and Crystallization of Polymers by Differential Scanning Calorimetry
ASTM D3951	-	Standard Practice for Commercial Packaging

(Copies of these documents are available online at https://www.astm.org/.)

UNDERWRITERS LABORATORIES

UL 94	-	Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 746B	-	Polymeric Materials - Long Term Property Evaluations

(Copies of these documents are available online at https://www.ul.com/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall consist of polyethersulfone thermoplastic molding or extrusion material, with or without fiber reinforcement or PTFE lubricant (see 6.3).

3.2 Properties of PES thermoplastic. Unless otherwise specified (see 6.2), the properties of the PES thermoplastic materials shall be as specified herein.

3.2.1 Nonreinforced PES. The properties of the nonreinforced PES thermoplastic molding and extrusion materials shall be as specified in table I.

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TABLE I. Properties of nonreinforced polyethersulfone. [1/]

Properties	Property Values	Test Method
General		
Melt index	See 1.2 and 4.5.1	ASTM D1238
Specific gravity	1.37 ± 0.02	ASTM D792
Mold shrinkage	0.007 in/in	ASTM D955
Water absorption, 24 hour max	0.5%	ASTM D570
Mechanical		
Tensile strength, yield, min [2/]	82 MPa (12,000 psi)	ASTM D 638
Tensile elongation, ultimate, min [2/]	40-80%	
Flexural strength, min [2/]	124 MPa (18,000 psi)	ASTM D790
Flexural modulus, min [2/]	2551 MPa (370,000 psi)	
Notched IZOD impact strength [3/, 4/]	80 J/m (1.5 ft-lb/in)	ASTM D256
Unnotched IZOD impact strength [3/, 4/]	No break	
Rockwell hardness, min	M80	ASTM D785
Thermal		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	200°C	ASTM D648
Glass transition temperature	222 ± 5°C	ASTM D3418
Coefficient of linear thermal expansion	3.1 x 10^{-5} in/in/°F	ASTM D696
Electrical		
Dielectric strength, min	400 volts/mil	ASTM D149
Dielectric constant at 1 MHz, max	3.5	ASTM D150
Dissipation factor at 1 MHz, max	0.0035	
Volume resistivity, min	10^{17} ohm-cm	ASTM D257
Flammability		
Limiting oxygen index, min	40	ASTM D2863
Flammability rating	94V-O	UL 94

[1/] All classes and grades of nonreinforced PES have the same minimum and maximum property requirements.
[2/] MPa x 145 = psi
[3/] J/m x 0.01873 = ft-lb/in
[4/] Alternative impact tests are contained in ASTM D3021. Additional impact tests and requirements shall be specified by the procuring activity.

3.2.2 Glass fiber reinforced PES. The properties of the glass fiber reinforced PES thermoplastic materials shall be as specified in tables IIa - IIe.

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TABLE IIa. Properties of class 1, 15% GFR polyethersulfone.

Properties	Property Values	Test Method
General		
Melt index	1.47 ± 0.02	ASTM D792
Specific gravity	15 ± 2%	4.6.1
Mold shrinkage	0.003 in/in	ASTM D955
Water absorption, 24 hour max	0.5%	ASTM D570
Mechanical		
Tensile strength, yield, min [1/]	114 MPa (16,500 psi)	ASTM D 638
Tensile elongation, ultimate, min [1/]	1-5%	
Flexural strength, min [1/]	158 MPa (23,000 psi)	ASTM D790
Flexural modulus, min [1/]	4,827 MPa (700,000 psi)	
Notched IZOD impact strength [2/, 3/]	---	ASTM D256
Unnotched IZOD impact strength [2/, 3/]	---	
Rockwell hardness, min	R118	ASTM D785
Thermal		
Thermal index, max	---	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	200°C	ASTM D648
Glass transition temperature	225 ± 5°C	ASTM D3418
Coefficient of linear thermal expansion	2.0 x 10^{-5} in/in/°F	ASTM D696
Electrical		
Dielectric strength, min	380 v/mil (14.9 Mv/m)	ASTM D149
Dielectric constant at 1 MHz, max	3.9	ASTM D150
Dissipation factor at 1 MHz, max	0.007	
Volume resistivity, min	10^{16} ohm-cm	ASTM D257
Flammability		
Limiting oxygen index, min	40	ASTM D2863
Flammability rating	94V-O	UL 94

[1/] MPa x 145 = psi
[2/] J/m x 0.01873 = ft-lb/in
[3/] Alternative impact tests are contained in ASTM D3021. Additional impact tests and requirements shall be specified by the procuring activity.

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TABLE IIb. Properties of class 2, 20% GFR polyethersulfone.

Properties	Property Values	Test Method
General		
Specific gravity	1.51 ± 0.02	ASTM D792
Glass fiber content by weight	20 ± 2%	4.6.1
Mold shrinkage	0.003 in/in	ASTM D955
Water absorption, 24 hour max	0.5%	ASTM D570
Mechanical		
Tensile strength, yield, min [1/]	103 MPa (15,000 psi)	ASTM D 638
Tensile elongation, ultimate, min [1/]	1-5%	
Flexural strength, min [1/]	158 MPa (23,000 psi)	ASTM D790
Flexural modulus, min [1/]	5,517 MPa (800,000 psi)	
Notched IZOD impact strength [2/, 3/]	75 J/m (1.4 ft-lb/in)	ASTM D256
Unnotched IZOD impact strength [2/, 3/]	427 J/m (8.0 ft-lb/in)	
Rockwell hardness, min	M98	ASTM D785
Thermal		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	205°C	ASTM D648
Glass transition temperature	225 ± 5°C	ASTM D3418
Coefficient of linear thermal expansion	2.0 x 10^{-5} in/in/°F	ASTM D696
Electrical		
Dielectric strength, min	380 v/mil (14.9 Mv/m)	ASTM D149
Dielectric constant at 1 MHz, max	3.9	ASTM D150
Dissipation factor at 1 MHz, max	0.007	
Volume resistivity, min	10^{16} ohm-cm	ASTM D257
Flammability		
Limiting oxygen index, min	40	ASTM D2863
Flammability rating	94V-O	UL 94

[1/] MPa x 145 = psi
[2/] J/m x 0.01873 = ft-lb/in
[3/] Alternative impact tests are contained in ASTM D3021. Additional impact tests and requirements shall be specified by the procuring activity.

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TABLE IIc. Properties of class 3, 30% GFR polyethersulfone.

Properties	Property Values	Test Method
General		
Melt index	1.60 ± 0.02	ASTM D792
Specific gravity	30 ± 2%	4.6.1
Mold shrinkage	0.002-0.003 in/in	ASTM D955
Water absorption, 24 hour max	0.5%	ASTM D570
Mechanical		
Tensile strength, yield, min [1/]	117 MPa (17,000 psi)	ASTM D 638
Tensile elongation, ultimate, min [1/]	1-5%	
Flexural strength, min [1/]	172 MPa (25,000 psi)	ASTM D790
Flexural modulus, min [1/]	7,586 MPa (1,100,000 psi)	
Notched IZOD impact strength [2/, 3/]	80 J/m (1.5 ft-lb/in)	ASTM D256
Unnotched IZOD impact strength [2/, 3/]	534 J/m (10 ft-lb/in)	
Rockwell hardness, min	M98	ASTM D785
Thermal		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	210°C	ASTM D648
Glass transition temperature	225 ± 5°C	ASTM D3418
Coefficient of linear thermal expansion	1.8 x 10^{-5} in/in/°F	ASTM D696
Electrical		
Dielectric strength, min	---	ASTM D149
Dielectric constant at 1 MHz, max	---	ASTM D150
Dissipation factor at 1 MHz, max	---	
Volume resistivity, min	10^{16} ohm-cm	ASTM D257
Flammability		
Limiting oxygen index, min	40	ASTM D2863
Flammability rating	94V-O	UL 94

[1/] MPa x 145 = psi
[2/] J/m x 0.01873 = ft-lb/in
[3/] Alternative impact tests are contained in ASTM D3021. Additional impact tests and requirements shall be specified by the procuring activity.

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TABLE IId. Properties of class 4, 40% GFR polyethersulfone.

Properties	Property Values	Test Method
General		
Melt index	1.68 ± 0.02	ASTM D792
Specific gravity	40 ± 2%	4.6.1
Mold shrinkage	0.0015 in/in	ASTM D955
Water absorption, 24 hour max	0.5%	ASTM D570
Mechanical		
Tensile strength, yield, min [1/]	131 MPa (19,000 psi)	ASTM D 638
Tensile elongation, ultimate, min [1/]	1-5%	ASTM D 638
Flexural strength, min [1/]	207 MPa (30,000 psi)	ASTM D790
Flexural modulus, min [1/]	11,034 MPa (1,600,000 psi)	ASTM D790
Notched IZOD impact strength [2/, 3/]	85 J/m (1.6 ft-lb/in)	ASTM D256
Unnotched IZOD impact strength [2/, 3/]	640 J/m (12 ft-lb/in)	
Rockwell hardness, min	M98	ASTM D785
Thermal		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	210°C	ASTM D648
Glass transition temperature	225 ± 5°C	ASTM D3418
Coefficient of linear thermal expansion	1.6 x 10^{-5} in/in/°F	ASTM D696
Electrical		
Dielectric strength, min	---	ASTM D149
Dielectric constant at 1 MHz, max	---	ASTM D150
Dissipation factor at 1 MHz, max	---	
Volume resistivity, min	10^{16} ohm-cm	ASTM D257
Flammability		
Limiting oxygen index, min	40	ASTM D2863
Flammability rating	94V-O	UL 94

[1/] MPa x 145 = psi
[2/] J/m x 0.01873 = ft-lb/in
[3/] Alternative impact tests are contained in ASTM D3021. Additional impact tests and requirements shall be specified by the procuring activity.

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TABLE IIe. Properties of class 5, 15% PTFE/30% GFR polyethersulfone.

Properties	Property Values	Test Method
General		
Melt index	1.70 ± 0.02	ASTM D792
Specific gravity	30 ± 2%	4.6.1
Mold shrinkage	0.0015 in/in	ASTM D955
Water absorption, 24 hour max	0.5%	ASTM D570
Mechanical		
Tensile strength, yield, min [1/]	103 MPa (15,000 psi)	ASTM D 638
Tensile elongation, ultimate, min [1/]	1-5%	ASTM D 638
Flexural strength, min [1/]	158 MPa (23,000 psi)	ASTM D790
Flexural modulus, min [1/]	5,517 MPa (800,000 psi)	ASTM D790
Notched IZOD impact strength [2/, 3/]	80 J/m (1.5 ft-lb/in)	ASTM D256
Unnotched IZOD impact strength [2/, 3/]	347 J/m (6.5 ft-lb/in)	
Rockwell hardness, min	M98	ASTM D785
Thermal		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	210°C	ASTM D648
Glass transition temperature	225 ± 5°C	ASTM D3418
Coefficient of linear thermal expansion	1.9 x 10^{-5} in/in/°F	ASTM D696
Electrical		
Dielectric strength, min	---	ASTM D149
Dielectric constant at 1 MHz, max	---	ASTM D150
Dissipation factor at 1 MHz, max	---	
Volume resistivity, min	10^{16} ohm-cm	ASTM D257
Flammability		
Limiting oxygen index, min	40	ASTM D2863
Flammability rating	94V-O	UL 94

[1/] MPa x 145 = psi
[2/] J/m x 0.01873 = ft-lb/in
[3/] Alternative impact tests are contained in ASTM D3021. Additional impact tests and requirements shall be specified by the procuring activity.

3.2.3 Carbon fiber reinforced PES. The properties of the carbon fiber reinforced PES thermoplastic materials shall be as specified in table IIIa and IIIb.

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TABLE IIIa. Properties of class 1, 30% CFR polyethersulfone.

Properties	Property Values	Test Method
General		
Specific gravity	1.48 ± 0.02	ASTM D792
Carbon fiber content by weight	30 ± 2%	ASTM D3171
Mold shrinkage	---	ASTM D955
Water absorption, 24 hour max	0.30%	ASTM D570
Mechanical		
Tensile strength, yield, min [1/]	158 MPa (23,000 psi)	ASTM D 638
Tensile elongation, ultimate, min [1/]	2-3%	
Flexural strength, min [1/]	234 MPa (34,000 psi)	ASTM D790
Flexural modulus, min [1/]	12,066 MPa (1,750,000 psi)	
Notched IZOD impact strength [2/]	53 J/m (1.0 ft-lb/in)	ASTM D256
Unnotched IZOD impact strength [2/]	267 J/m (5.0 ft-lb/in)	
Thermal		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	210°C	ASTM D648
Coefficient of linear thermal expansion	2.0 x 10^{-5} in/in/°F	ASTM D696
Flammability		
Flammability rating	94V-O	UL 94

[1/] MPa x 145 = psi
[2/] J/m x 0.01873 = ft-lb/in

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TABLE IIIb. Properties of class 2, 15% PTFE/30% CFR polyethersulfone.

Properties	Property Values	Test Method
General		
Specific gravity	1.57 ± 0.02	ASTM D792
Carbon fiber content by weight	30 ± 2%	ASTM D3171
Mold shrinkage	0.001 - 0.003 in/in	ASTM D955
Water absorption, 24 hour max	0.25%	ASTM D570
Mechanical		
Tensile strength, yield, min 1/	128 MPa (20,000 psi)	ASTM D 638
Tensile elongation, ultimate, min 1/	2 - 3%	
Flexural strength, min 1/	214 MPa (31,000 psi)	ASTM D790
Flexural modulus, min 1/	11,720 MPa (1,700,000 psi)	
Notched IZOD impact strength 2/	43 J/m (0.8 ft-lb/in)	ASTM D256
Unnotched IZOD impact strength 2/	214 J/m (4.0 ft-lb/in)	
Thermal		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	210°C	ASTM D648
Coefficient of linear thermal expansion	0.8 x 10^{-5} in/in/°F	ASTM D696
Flammability		
Flammability rating	---	UL 94

1/ MPa x 145 = psi
2/ J/m x 0.01873 = ft-lb/in

3.3 Form. The form of the PES thermoplastic resins shall be granules, pellets, or powder. When specified by the procuring activity, the form shall be in the size and shape specified (see 6.2).

3.4 Workmanship. The material shall be free of contamination.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of tests. All quality conformance tests shall be classified, as follows:

a. Lot acceptance tests (see 4.5).
b. Periodic lot check tests (see 4.6).

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4.3 Lot size. Lot size shall consist of all the molding or extrusion material manufactured at one time, from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.4 Sampling. A sufficient quantity of material from one unit representative of each lot (4.3) shall be selected at random for quality conformance testing. Each container of material shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.5 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot, except when periodic lot cheek tests are also required. Lot acceptance tests shall consist of the tests specified herein.

4.5.1 Melt flow. Melt flow shall be determined for nonreinforced PES materials only, in accordance with ASTM D 1238 using a load of 5000 g (including push rod) at a temperature of 360 ± 5°C.

4.5.2 Glass transition temperature. The glass transition temperature, Tg, shall be determined in accordance with ASTM D 3418.

4.5.3 Tensile strength. Tensile strength shall be determined in accordance with ASTM D 638.

4.5.4 Flammability. Unless otherwise specified (see 6.2 and 6.4), oxygen index and flammability rating tests shall be determined in accordance with ASTM D 2863 and UL 94, respectively.

4.6 Periodic lot check tests. Periodic lot check tests shall be made on a lot of material as specified by the procuring activity (see 6.2). Periodic lot check tests shall consist of all the test methods specified in tables I, II and III (except for those test methods which are classified as lot acceptance tests in 4.5.1, 4.5.2, 4.5.3 and 4.5.4) and the additional test methods specified herein. When periodic lot check tests are made, they shall be in the basis for acceptance or rejection of the lot.

4.6.1 Fiberglass content (type II only). The following procedure shall be used to determine the fiberglass content of the PES material:

a. Test five samples. Each sample shall have a minimum weight of 3 grams and shall be in a form suitable for weighing in an ignition crucible. Condition the sample in a previously weighed; ignited crucible in a desiccator for a minimum of 8 hours. Weigh the conditioned sample and crucible to within ± 0.005 gram.

b. Place the weighed crucibles with the test samples in a muffle furnace and raise the furnace temperature to 565 ± 10°C (1050 ± 50°F) at a rate that will not cause blowing or loss of the glass reinforcement. Heat the samples at 565 ± 10°C (1050 ± 50°F) to constant weight. Cool the ignited crucibles in a desiccator. Weigh to within ± 0.005 gram.

c. Calculate the glass content as follows:

$$\text{Glass content, \% (by weight)} = \frac{\text{Original weight - Weight loss}}{\text{Original weight}} \times 100$$

4.6.2 Impact resistance. Impact resistance shall be determined in accordance with ASTM D3029.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1 Packaging notes. Due to the substantial passage of time since the last revision of this specification, additional packaging information that may be helpful including past definitions and procedures for unit of product, sampling, packaging inspection, and exterior container markings are included in 6.5 and its sections therein. This legacy content will be deleted in the next revision or amendment to this specification.

5.2 Marking. Unless otherwise indicated in the contract or purchase order, any special marking required by shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The polyethersulfone (PES) materials covered by this specification are engineering thermoplastics intended for continuous use at 180°C. This thermoplastic is intended for use in various aircraft, aerospace and military applications, such as instrumentation housings, fittings, electrical connectors, switches terminal blocks, nose cones, radomes, air ducting, obturator bands, and as a matrix for glass fiber and carbon-graphite composites. It works well in extrusions, injection molding, and also as feedstock material for additive manufacturing.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Type and class of PES material
c. Properties of PES material, if other than specified in 3.2
d. Specific form, if other than specified in 3.3
e. Flammability tests, if other than specified in 4.5.4
f. Periodic lot check tests, if required (see 4.6)
g. Level of packing required (see 5.1)

6.3 Polyether sulfones. Polyether sulfones are high temperature engineering thermoplastics that contain sulfone and ether groups in the polymer chain. All commercial products have phenylene rings in the polymer chain and are derived from poly(sulfo-1, 4-phenylene-oxy-1, 4-phenylene). Poly(p-phenylene) sulfone has a melting temperature of about 520°C.

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6.4 Flammability test (aircraft). For aircraft applications, flammability testing in accordance with Federal Aviation Regulation 25, paragraph 853 may be required.

6.5 Packaging considerations. Due to the substantial length of time since the last revision of this specification, additional packaging information including past definitions and procedures for unit of product, sampling, and packaging inspection are included as a courtesy in this section and procuring officials should follow current procedures for packaging in 5.1.

6.5.1 Packing. Packing should conform to level A or industrial as specified.

6.5.1.1 Level A. Unless otherwise specified, the material should be packed in metal drums conforming to PPP-D-729, type III or type IV, in quantities of 181 kg (400 lbs), maximum.

Insofar as practical, drums should be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums should contain identical quantities and should be closed in accordance with the applicable container specification. Fiber drums should be furnished with a 4 mil thick polyethylene liner properly heat sealed.

6.5.1.2 Industrial. The material should be packed in accordance with ASTM D 3951.

6.6 Subject term (key word) listing.

AM feedstock	Fiber Reinforced	PES
Extrusion	Injection molding	PTFE

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - EC
Air Force - 20

Preparing activity:
ARMY - MR

(Project 9330-2026-002)

Review activities:

Army - GL, SM
Navy - AS, CG, SH
Air Force - 03, 84
DLA - GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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